ENHANCED OIL RECOVERY METHOD SELECTION

INTRODUCTION

- Enhanced Oil Recovery (EOR) Techniques Are Essential For Improving Oil Production From Mature Reservoirs Where Conventional Methods Are Insufficient. Selecting An Appropriate EOR Method Is A Complex Task Influenced By Multiple Reservoir And Fluid Parameters.
- Traditional Approaches Rely On Expert Judgment And Empirical Rules, Which Can Be Time-consuming And Subjective. With The Advancement Of Artificial Intelligence, Machine Learning Offers An Efficient Data-driven Solution For Automated And Accurate EOR Method Selection.

LITERATURE REVIEW

TABER ET AL.-

Developed Screening Guidelines For Eor Method Selection Using Reservoir Parameters

MOHAGHEGH ET AL.-

Demonstrated The Use Of Neural Networks In Petroleum Engineering Applications.

ML-BASED APPROACHES

Classification Models Such As Decision Trees And Random Forests Improve Accuracy In Eor Method Prediction

LIMITATION-

Existing

Methods Lack Generalization And Rely On Limited Data

RESEARCH GAP:Need for a robust AI-based decision support system for accurate EOR method selection

OBJECTIVES

1. To Design An Intelligent Decision Support System For Automated Selection Of Appropriate Enhanced Oil Recovery Methods Based On Reservoir And Fluid Characteristics.

2. To Develop a Robust Machine Learning Classification Framework That Learns Complex, Non-linear Relationships Between Reservoir Parameters And Suitable EOR Techniques.

3. To Generate And Prepare A High-Quality Reservoir Dataset By Combining Domain Knowledge With Data Preprocessing And Feature Engineering Techniques.

4. To compare Multiple Machine Leraning Algorithm And Identify The Most Accurate And Reliable Model For EOR Method Prediction.

5. To Improve The Consistency And Objectivity Of EOR Method Selection Compared To Traditional Rule-based And Experience-driven Approaches.

DATASET

MODEL 1 (RANDOM FOREST)	MODEL 2(ARTIFICIAL NEURAL NETWORK)
Source: Kaggle And Uci Machine Learning Repositories. **Data Type:** Tabular Reservoir Data With Geological And Fluid Parameters. **Features:** Depth, Porosity, Permeability, Temperature, Pressure, Oil Viscosity, Api Gravity. Number Of Samples: 800+ Reservoir Instances. **Preprocessing:** Missing Value Handling, Outlier Removal, And Feature Normalization. **Dataset Split:** 70% Training, 30% Testing. **Label:** Eor Method Class (Thermal, Gas, Chemical, Waterflood). **Purpose:** Training A Robust Tree-based Classifier For Eor Method Selection.	**Source:** Synthetic Data Generated Based On Petroleum Engineering Literature. **Data Type:** Normalized Tabular Data Suitable For Neural Network Training. **Features:** Depth, Porosity, Permeability, Temperature, Pressure, Oil Viscosity, Api Gravity. Number Of Samples: 1200+ Synthetic Reservoir Instances. **Preprocessing:** Feature Scaling, Min–max Normalization, And One-hot Encoding Of Target Classes. **Dataset Split:** 70% Training, 15% Validation, 15% Testing. **Purpose:** Training A Multi-layer Neural Network For Non-linear Pattern Learning.

ARCHITECTURE

RESERVOIR DATASET

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DATA PREPROCESSING

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FEATURE ENGINEERING

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TRAIN–TEST SPLIT

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MODEL TRAINING

(RANDOM FOREST / ANN)

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MODEL EVALUATION

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EOR METHOD PREDICTION

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DECISION SUPPORT OUTPUT

1.Input Data Layer

1. Reservoir and fluid parameters
2. (Depth, Porosity, Permeability, Temperature, Pressure, Oil Viscosity, API Gravity)

2.Data Preprocessing Module

1. Missing value handling 2.Outlier removal 3.Normalization / Scaling

3.Feature Engineering Module

1. Feature selection
2. Encoding target classes

4.Dataset Split Module

1. Training Set 2.Validation Set (for ANN) 3.Testing Set

5.Model Layer

1. Random Forest Classifier
2. Artificial Neural Network

6.Model Evaluation Module

1. Accuracy 2.Precision 3.Recall 4.F1-score

7.Prediction Module

1. Predicted EOR Method
2. (Thermal / Gas / Chemical / Waterflood)

8.Decision Support Interface

1. Final recommended EOR technique

SYSTEM ARCHITECTURE

- The Proposed System Takes Reservoir And Fluid Parameters As Input And Performs Data Preprocessing And Feature Engineering To Prepare A Clean Dataset. The Processed Data Is Then Split Into Training And Testing Sets. Two Machine Learning Models, Random Forest And Artificial Neural Network, Are Trained Independently. The Trained Models Are Evaluated Using Standard Performance Metrics. Finally, The System Predicts The Most Suitable Enhanced Oil Recovery Method And Provides A Decision Support Output For Engineers.

CONCLUSION

- This Work Proposed An Ai-based Framework For Automated Enhanced Oil Recovery Method Selection Using Random Forest And Artificial Neural Network Models. The Results Show That Machine Learning Can Effectively Model Complex Reservoir Behavior And Improve Decision Accuracy. The Proposed System Provides A Practical Decision Support Tool For Petroleum Engineers. Future Work Can Extend This Framework Using Larger Datasets And Advanced Deep Learning Techniques.